INCH - POUND

MIL-N-47187B(MI)
31 August 1990
SUPERSEDING
MIL-N-47187A(MI)
18 January 1984

MILITARY SPECIFICATION

NUTS, BLIND, PRESS, AND BLIND RIVET, INSTALLATION OF

This specification is approved for use by the U.S. Army Missile Command, Department of the Army, and is available for use by all Departments and Agencies of the Department of Defense.

1. SCOPE

1.1 Scope. This specification covers the requirements for the installation of blind nuts, press nuts, and blind rivet nuts.

2. APPLICABLE DOCUMENTS

2.1 Government documents.

2.1.1 Specifications, standards, and handbooks. The following specifications, standards, and handbooks form a part of this document to the extent specified herein. Unless otherwise specified, the issues of these documents are those listed in the issue of the Department of Defense Index of Specifications and Standards (DODISS) and supplement thereto, cited in the solicitation (see 6.2).

SPECIFICATIONS

FEDERAL

TT-P-1757 - Primer Coating, Zinc Chromate, Low-Moisture-Sensitivity

Beneficial comments (recommendations, additions, deletions) and any pertinent data which may be of use in improving this document should be addressed to: Commander, U.S. Army Missile Command, ATTN: AMSMI-RD-SE-TD-ST, Redstone Arsenal, AL 35898-5270 by using the Standardization Document Improvement Proposal (DD Form 1426) appearing at the end of this document or by letter.

AMSC N/A FSC 14GP

DISTRIBUTION STATEMENT A. Approved for public release; distribution is unlimited.

Source: https://assist.dla.mil -- Downloaded: 202640-22T07:57Z
Check the source to verify that this is the current version before use.

(Unless otherwise indicated, copies of the federal and military specifications, standards, and handbooks are available from the Standardization Documents Order Desk, Bldg. 4D, 700 Robbins Ave., Philadelphia, PA 19111-5094.)

2.2 Non-Government publications. The following document(s) form a part of this document to the extent specified herein. Unless otherwise specified, the issues of the documents which are DoD adopted are those listed in the issue of the DODISS cited in the solicitation. Unless otherwise specified, the issues of documents not listed in the DODISS are the issues of the documents cited in the solicitation (see 6.2).

2-1008 - HI-Shear Blind Nut Data Book

(Application for copies should be addressed to the HI-Shear Industries, Inc., 2600-T Skypark Drive, Torrance, CA 90509.)

(Non-Government standards and other publications are normally available from the organizations that prepare or distribute the documents. These documents also may be available in or through libraries or other informational services.)

2.3 Order of precedence. In the event of a conflict between the text of this document and the references cited herein, the text of this document takes precedence. Nothing in this document, however, supersedes applicable laws and regulations unless a specific exemption has been obtained.

3. REQUIREMENTS

3.1 Hole preparation. Holes shall be straight and perpendicular to the parent material and shall conform to the dimensional requirements specified on the engineering drawing and herein.

3.2 Installation preparation. All surface treatments such as anodize or chemical film treat shall have been completed on the parent material prior to installation. The blind-nut sleeves and press nuts shall be installed with wet zinc chromate primer conforming to TT-P-1757, color Y.

3.3 Installation of blind nuts. Installation of blind nuts shall be as specified in figure 1. Installation shall be such that the sleeve is flush with or not greater than 0.015 inch below the structure surface.

3.3.1 Installation force. Blind nuts shall be installed as shown in figure 2 by engaging the threads of the expander and pulling the expander into the sleeve with the force specified in table I.

3.3.2 Hole size and configuration. Hole size shall be as specified in table II. Configuration of the holes shall be as specified in figure 3.

Source: https://assist.dla.mil -- Downloaded: 2026-10-22T07:57Z
Check the source to verify that this is the current version before use.

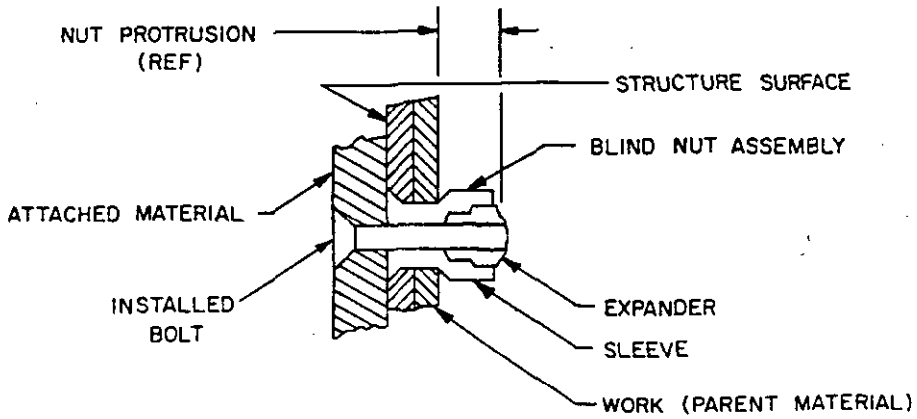


FIGURE 1. Blind nut components and installation.

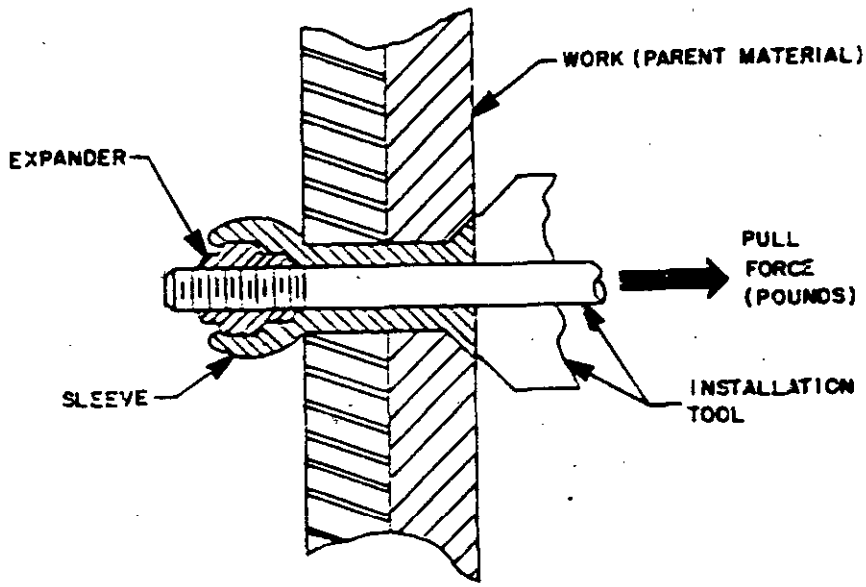


FIGURE 2. Blind nut installation tool.

Source: https://assist.dla.mil -- Downloaded: 2026-10-22T07:57Z
Check the source to verify that this is the current version before use.

TABLE I. Blind nut installation pull force.

Thread size	Pull force (pounds) 1/
4-40UNC-3B	775-792
6-32UNC-3B	1425-1495
8-32UNC-3B	2550-2680
10-32UNF-3B	3875-4050
1/4-28UNF-3B	6100-6400
5/16-24UNF-3B	8125-8540
3/8-24UNF-3B	9700-10,200
1/2-20UNF-3B	9700-10,200

1/ Installation pull force shall be applied using equipment as specified in HI-Shear Blind Nut Data Book 2-1008, Revised 23 January 1968, HI-Shear Corp, Torrance, California; or equal.

TABLE II. Blind nut installation hole requirements.

Blind nut thread	Hole size D diameter (inch)	Sleeve countersink C diameter (inch) (ref)	
		Shallow head 1/	Shear head 2/
4-40UNC-3B	0.170-0.173	0.229	-
6-32UNC-3B	0.219-0.222	0.278	-
8-32UNC-3B	0.258-0.264	0.317	0.395
10-32UNF-3B	0.313-0.319	0.372	0.474
1/4-28UNF-3B	0.391-0.397	0.450	0.560
5/16-24UNF-3B	0.469-0.477	0.528	0.668
3/8-24UNF-3B	0.531-0.539	0.590	0.754
1/2-20UNF-3B	0.656-0.664	0.714	-

1/ Type BN360, HI-Shear Corp, Torrance, California; or equal.

2/ Type BN359, HI-Shear Corp. Torrance, California; or equal.

Source: https://assist.dla.mil -- Downloaded: 2026-10-22T07:57Z
Check the source to verify that this is the current version before use.

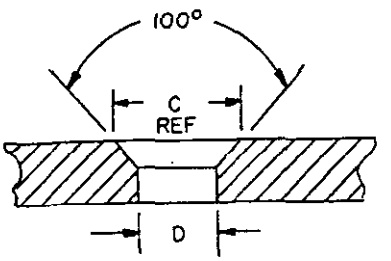


FIGURE 3. Blind nut hole configuration.

3.4 Installation of press nuts. Installation requirements for press nuts shall be as specified in table III. Press nuts shall be positioned with pilot flange in hole and force shall be applied of sufficient magnitude to cause the flange of the press nut to bear flush against the surface of the parent material (see figure 4.)

TABLE III. Press nut installation hole requirements.

Press nut thread size		Hole diameter (inch) 1/
Coarse	Fine	
4-40NC-3B	-	0.172-0.175
6-32NC-3B	-	0.191-0.194
8-32NC-3B	-	0.221-0.224
-	10-32NF-3B	0.250-0.253
-	1/4-28UNF-3B	0.344-0.347
-	5/16-24UNF-3B	0.413-0.418
-	3/8-24UNF-3B	0.484-0.489

1/ Type HRPN, Rosan Inc., Newport Beach, California; or equal.

Source: https://assist.dla.mil -- Downloaded: 202640-22T07:57Z
Check the source to verify that this is the currentversion before use.

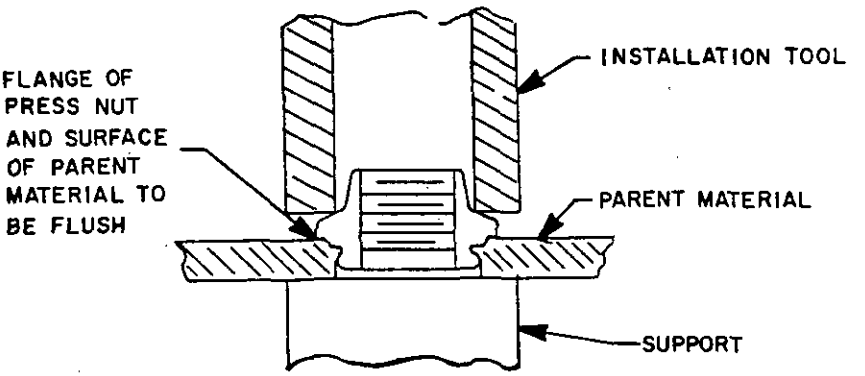


FIGURE 4. Press nut installation.

3.5 Installation of blind rivet nuts. Installation of blind rivet nuts having countersunk heads or flat heads and keyed or keyless configurations shall be as specified in figure 5. Installation of blind rivet nuts having countersunk heads shall be such that the head is flush with or under flush to the structure surface.

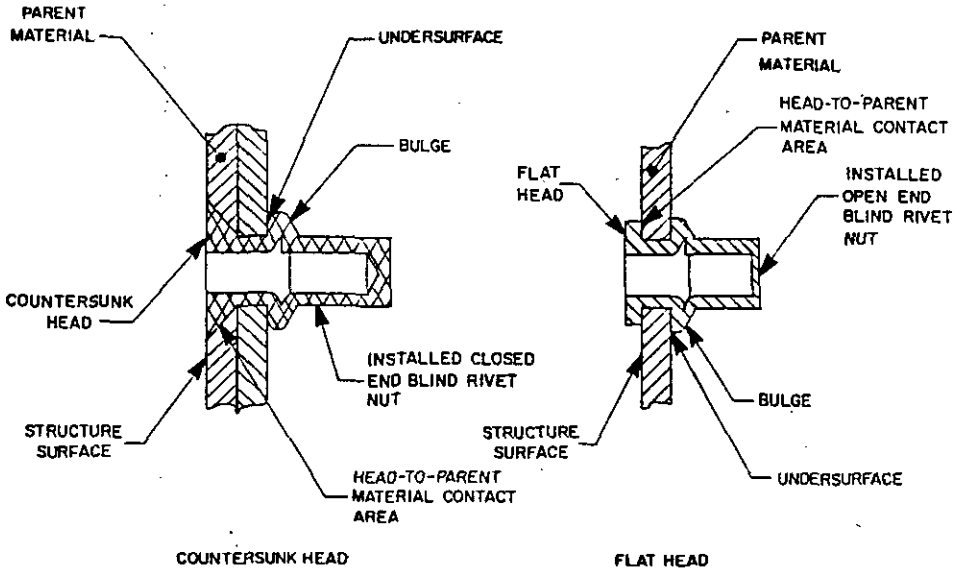


FIGURE 5. Blind rivet nut installation.

3.5.1 Head-to-parent material contact. At least 67 percent of the head-to-parent material contact area (see figure 5) of the blind rivet nut head shall be in contact with the parent material of the structure.

3.5.2 Bulge and tightness. The blind rivet nut shall be firmly secured in the parent material of the structure so there is no relative movement between the installed insert and the structure. The undersurface of the structure (see figure 5) shall be flat so the bulge of the installed blind rivet nut will form normally.

Source: https://assist.dla.mil -- Downloaded: 202640-22T07:57Z
Check the source to verify that this is the currentversion before use.

3.5.3 Hole size and configuration. Configuration of the holes shall be as specified in figure 6. When a keyway is required, it shall be cut with a key cutter tool or guided drill. Hole size shall be as specified in table IV.

3.5.4 Replacement of blind rivet nuts. Replacement shall be made with the same diameter and configuration, in accordance with the installation procedure of this specification.

3.6 Workmanship. Blind nuts, press nuts, and blind rivet nuts shall be uniformly installed and shall reflect a high quality of workmanship.

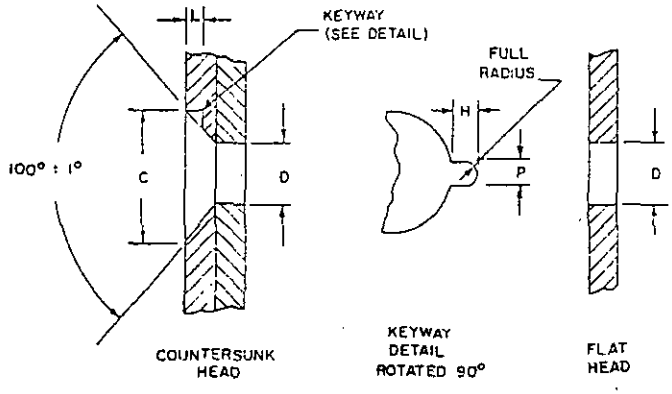


FIGURE 6. Blind rivet nut hole configuration.

4. QUALITY ASSURANCE PROVISIONS

4.1 Responsibility for inspection. Unless otherwise specified in the contract or purchase order, the contractor is responsible for the performance of all inspection requirements (examinations and tests) as specified herein. Except as otherwise specified in the contract or purchase order, the contractor may use his own or any other facilities suitable for the performance of the inspection requirements specified herein, unless disapproved by the Government. The Government reserves the right to perform any of the inspections set forth in this specification where such inspections are deemed necessary to ensure supplies and services conform to prescribed requirements.

4.1.1 Responsibility for compliance. All items shall meet all requirements of sections 3 and 5. The inspection set forth in this specification shall become a part of the contractor's overall inspection system or quality program. The absence of any inspection requirements in the specification shall not relieve the contractor of the responsibility of ensuring that all products or supplies submitted to the Government for acceptance comply with all requirements of the contract. Sampling inspection, as part of manufacturing operations, is an acceptable practice to ascertain conformance to requirements, however, this does not authorize submission of known defective material, either indicated or actual, nor does it commit the Government to accept defective material.

Source: https://assist.dla.mil -- Downloaded: 2026-10-22T07:57Z
Check the source to verify that this is the current version before use.

TABLE IV. Blind rivet nut installation hole requirements.

Blind rivet nut thread	Hole size D diameter (inch)	Countersink C diameter (inch)	Keyway dimensions		
			Width P (inch)	Height H (inch)	Length L (inch)
4-40UNC-3B	0.155-0.157	0.263	0.062-0.065	0.046-0.048	0.051 min.
6-32UNC-3B	0.189-0.193	0.323	0.062-0.065	0.056-0.058	0.063 min.
8-32UNC-3B	0.221-0.226	0.355	0.062-0.065	0.056-0.058	0.063 min.
10-32UNF-3B	0.250-0.256	0.391	0.062-0.065	0.056-0.058	0.065 min.
1/4-20UNC-3B	0.332-0.338	0.529	0.062-0.065	0.056-0.058	0.089 min.
5/16-18UNC-3B	0.413-0.423	0.656	0.128-0.131	0.097-0.102	0.104 min.
3/8-16UNC-3B	0.490-0.500	0.777	0.128-0.131	0.110-0.115	0.124 min.

Source: https://assist.dla.mil -- Downloaded: 2026-10-22T07:57Z
Check the source to verify that this is the current version before use.

4.2 Classification of inspection. The inspection requirements specified herein are classified as quality conformance inspection (see 4.4).

4.3 Inspection conditions. Unless otherwise specified, all inspections shall be performed in accordance with the test conditions specified in 4.4.

4.4 Quality conformance inspection. Quality conformance inspection shall be as specified in table V.

TABLE V. Quality conformance inspection.

Inspection	Requirement paragraph	Test paragraph
Pre-installation	3.2	4.5.1
Blind nut installation	3.3	4.5.2
Installation force	3.3.1	4.5.2.1
Blind nut hole size and configuration	3.3.2	4.5.3
Press nut installation	3.4	4.5.4
Blind rivet nut installation	3.5	4.5.5
Head-to-parent material contact	3.5.1	4.5.5.1
Bulge and tightness	3.5.2	4.5.5.2
Blind rivet nut hole size and configuration	3.5.3	4.5.6
Workmanship	3.6	4.4.1

4.4.1 Inspection. Dimensional and visual inspections of samples shall be performed to determine compliance with the requirements specified in Section 3.

4.5 Test methods.

4.5.1 Chemical composition. Chemicals used for surface treatments and material prior to installation shall be in accordance with TT-P-1757 (see 3.1).

4.5.2 Blind nut installation test. Blind nut installation shall meet the requirements as specified in 3.2.

4.5.2.1 Installation force. Upon installation, the applied force shall be as specified in table I and 3.2.1.

4.5.3 Blind nut hole size and configuration. Blind nut installation hole requirements shall be as specified in table II and 3.2.2.

4.5.4 Press nut installation test. Press nut installation shall meet the requirements as specified in table III and 3.3.

4.5.5 Blind rivet nut installation test. Blind rivet nut installation shall be as specified in figure 5 and 3.4.

Source: https://assist.dla.mil -- Downloaded: 2026-10-22T07:57Z
Check the source to verify that this is the current version before use.

4.5.5.1 Head-to-parent material contact. The head-to-parent material contact area shall be as specified in figure 5 and 3.4.1.

4.5.5.2 Bulge and tightness. The bulge and tightness of the blind rivet nut shall be as specified in 3.5.2 and figure 5.

4.5.6 Blind rivet nut hole size and configuration. Hole size and configuration shall be as specified in table IV and 3.4.3.

5. PACKAGING

Section 5 is not applicable to this specification.

6. NOTES

(This section contains information of a general or explanatory nature that may be helpful, but is not mandatory.)

6.1 Intended use. This specification covers the requirements for the installation of blind nuts, press nuts, and blind rivet nuts.

6.2 Acquisition requirements. Acquisition documents must specify the following:

a. Title, number, and date of the specification.

b. Issue of DODISS to be cited in the solicitation, and if required, the specific issue of individual documents referenced (see 2.1.1 and 2.2).

6.3 Metrication. Metric equivalents in accordance with FED-STD-376 are acceptable for use in this specification.

6.4 Subject term (keyword) listing.

Anchor, sleeve
Fasteners, threaded
Lock washers

6.5 Changes from previous issue. Marginal notations are not used in this revision to identify changes with respect to the previous issue due to the extensiveness of the changes.

Custodian:
Army - MI

Preparing Activity
Army - MI

Project No. 14GP-A123

Source: https://assist.dla.mil -- Downloaded: 2026-10-22T07:57Z
Check the source to verify that this is the current version before use.

STANDARDIZATION DOCUMENT IMPROVEMENT PROPOSAL

INSTRUCTIONS

1. The preparing activity must complete blocks 1, 2, 3, and 8. In block 1, both the document number and revision letter should be given.
2. The submitter of this form must complete blocks 4, 5, 6, and 7.
3. The preparing activity must provide a reply within 30 days from receipt of the form.

NOTE: This form may not be used to request copies of documents, nor to request waivers, or clarification of requirements on current contracts. Comments submitted on this form do not constitute or imply authorization to waive any portion of the referenced document(s) or to amend contractual requirements.

I RECOMMEND A CHANGE:	1. DOCUMENT NUMBER MIL-N-47187B (MI)	2. DOCUMENT DATE (YYMMDD)

3. DOCUMENT TITLE
NUTS, BLIND, PRESS, AND BLIND RIVET, INSTALLATION OF

4. NATURE OF CHANGE *(Identify paragraph number and include proposed rewrite, if possible. Attach extra sheets as needed.)*

5. REASON FOR RECOMMENDATION

6. SUBMITTER

a. NAME (Last, First, Middle Initial)	b. ORGANIZATION	
c. ADDRESS (Include Zip Code)	d. TELEPHONE (Include Area Code) (1) Commercial (2) AUTOVON (If applicable)	7. DATE SUBMITTED (YYMMDD)

8. PREPARING ACTIVITY

a. NAME	b. TELEPHONE (Include Area Code) (1) Commercial (2) AUTOVON
c. ADDRESS (Include Zip Code)	IF YOU DO NOT RECEIVE A REPLY WITHIN 45 DAYS, CONTACT: Defense Quality and Standardization Office 5203 Leesburg Pike, Suite 1403, Falls Church, VA 22041-3466 Telephone (703) 756-2340 AUTOVON 289-2340

DD FORM 1426 89 OCT

Source: https://assist.dla.mil -- Downloaded: 2026-10-22T07:57Z
Check the source to verify that this is the current version before use.